Building the detector protection system

Hardware plan for the I&I workshop

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Draft revised 27 October 2026



Fermilab

Use a commercial PLC for DPS

Proposal: use the bottom drift electronics (BDE) PLC design and add the inputs and outputs needed by the other systems.

A programmable logic controller (PLC) applies the protection rules. Nearby I/O modules connect sensors and equipment.

Use OPC UA for monitoring and agreed supervisory commands.

Detector Protection System (DPS). Confirm the hardware choice after checking requirements and testing the prototype.

BDE protection and rack PDUs

Hardware	What we have	Next check
BDE protection	PLC design documents, slides and Ignition front end available	With Terri and Trevor, identify what to reuse and what DPS needs to add
Distributed I/O	BDE panels distributed along the mezzanine	Use the BDE layout to size and place additional DPS I/O
Rack power	Power distribution units (PDUs) are planned	Confirm models, sensors, trip logic and external signals

Based on the 25–26 October discussion and working hardware inventory. This is not an installed-equipment survey.

Put I/O near the equipment it protects

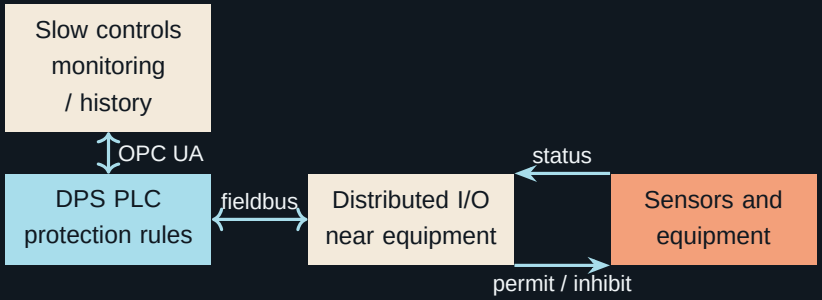


Diagram. Proposed structure; confirm each connection, protocol and electrical interface.

Decide how many PLCs we need and where to place them.

BDE uses an EtherCAT protection connection. Its extension, capacity and timing remain to be tested.

Check what each rack PDU can do locally

A local interlock acts at the equipment, without waiting for the central DPS.

1

Check the device

Which sensors can trigger a trip? Which outlets switch off? What happens if its network connection fails?

2

Check its DPS interface

Can it accept an external permit and report a trip? How do we reset it? Which status is available over SNMP?

Do not assume smoke, water or temperature sensing implies automatic shutdown. Confirm the model, configuration and wiring.

List the signals and count the I/O

Record for each path	What we need to choose
Source, destination, location	Panel layout and cable routes
Signal type, range, protocol	Digital / analog I/O, isolation and gateways
Channel / variable counts	Capacity, network load and spares
Required action and total time	PLC, local interlock or dedicated connection
Lost signal and recovery	Wiring, feedback and reset controls

Use the matrix to check interface control documents (ICDs). Remove overlaps; add CALCI and missing systems.

Confirm response times before choosing hardware

So far, I have not found a critical protection function that needs a response below 1 ms.

Each consortium must check the time from detecting a fault to completing the protective action.

Test sensor, communication, PLC and equipment delays together.

The consortia need to confirm this before procurement. Use a faster local interlock wherever the required response time demands it.

Check how DPS connects to the other systems

Boundary	Work needed
Slow controls (SC) / DPS	With Geoff, agree command priority, trip reset and re-enable; SC logs relevant data
Cryogenics / instrumentation	Find the actual source of each measurement and its path to DPS
LBNF/DUNE BMS / SDSTA BMS	Keep both building systems distinct; agree limited hardware I/O and where it connects
Networks / OPC UA	Agree monitoring, permitted commands and access across networks

Personnel-safety alarms remain with SDSTA. Software monitoring links do not establish a qualified protection path.

Survey the facility interfaces at SURF

1

Prepared

Marked inspection maps for levels 4850 and 4910. Coordinating access and safety requirements with Rachel.

2

Find on site

At the underground locations and Ross Dry: installed controllers, I/O, panel space and possible connection points.

Ask Dale Curran at SDSTA about the controllers and available connections.

Inspection planned for this week, subject to access and safety arrangements. Maps have not yet been shared; findings are pending.

Test the PLC, remote I/O and equipment together

Proposed bench tests	What to measure or check
PLC + remote I/O + representative load	Complete trip time under expected load
Rack PDU + sensors + DPS interface	Local shutdown, trip feedback and reset
OPC UA + SC supervisory commands	Command priority, logging and recovery
Signal, fieldbus, network and power loss	Defined equipment state and controlled restart

Use the simulation framework to prepare cases, then test on hardware. Agree the Fermilab bench scope and any SURF tests, hosts and equipment.

Design and test the hardware before June 2027

Proposed period	Deliverable
Now–December 2026	Review BDE; survey interfaces; list I/O and timing requirements
January–March 2027	Design PLC topology, panels, cabling and networks; estimate cost and lead times
April–June 2027	Test prototypes; document results; review design and procurement readiness
After review	Order approved hardware; schedule installation with the site teams

Check summer 2027 interface needs now. Target the full vertical drift (VD) response-matrix review for early 2028.

Agree who does the work

Proposed team	Next action
Manuel + process control engineer	Design the hardware; prepare specifications and purchases
Terri, Trevor + Manuel	Review BDE implementation and what DPS must add
Manuel + consortium contacts	Confirm signals, timing and protection in each project phase
Manuel + Geoff	Agree SC/DPS control and reset behaviour
Vivian + subsystem / DPS teams	Update ICD connections and bench-test scope

Confirm owners at I&I. When DPS is unavailable, propose that each consortium protects its equipment; agree this for every phase.